

Day_3_ Deu-Joz

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[]: DAY-03 Deutsch-Jozsa Algorithm in Qiskit: RAJIV KUMAR YADAV

Deutsch-Jozsa Algorithm in Qiskit

The Deutsch-Jozsa algorithm is a foundational quantum algorithm designed to determine whether a hidden Boolean function (an “oracle”) is constant (outputs the same value for all inputs) or balanced (outputs 0 for half the inputs and 1 for the other half). While a classical computer requires up to $(2^{n-1} + 1)$

Constant: same output for every input. Balanced: outputs 0 for exactly half the inputs and 1 for the other half.

The goal is to determine whether (f) is constant or balanced.

Deutsch-Jozsa demonstrates an exponential query-complexity advantage under a promise problem. It does not mean that a quantum computer is automatically (2^n) times faster for every real-world computation.

The advantage comes from combining:

Superposition + Phase Kickback + Interference

```
[2]: from qiskit import QuantumCircuit
      from qiskit_aer import AerSimulator
```

```
[3]: def deutsch_jozsa(n, oracle_type="balanced"):
      # n input qubits + 1 output qubit
      qc = QuantumCircuit(n + 1, n)

      # Put output qubit in |1>
      qc.x(n)

      # Apply Hadamard to all qubits
      qc.h(range(n + 1))

      # Oracle
      if oracle_type == "constant":
          # f(x) = 0
          pass

      elif oracle_type == "balanced":
```

```

    # Example balanced function:
    #  $f(x) = x_0 \text{ XOR } x_1 \text{ XOR } \dots \text{ XOR } x_{(n-1)}$ 
    for i in range(n):
        qc.cx(i, n)

    # Hadamard on input qubits
    qc.h(range(n))

    # Measure input qubits
    qc.measure(range(n), range(n))

    return qc

```

```

[6]: # Example: 3 input qubits
qc = deutsch_jozsa(3, "balanced")
print(qc)

```

```

q_0:  H      H      M

q_1:  H      H      M

q_2:  H      H      M

q_3:  X  H  X  X  X

c: 3/

                                0      1  2

```

```

[7]: # Run simulation
simulator = AerSimulator()
result = simulator.run(qc, shots=1024).result()

print(result.get_counts())

```

```

{'111': 1024}

```